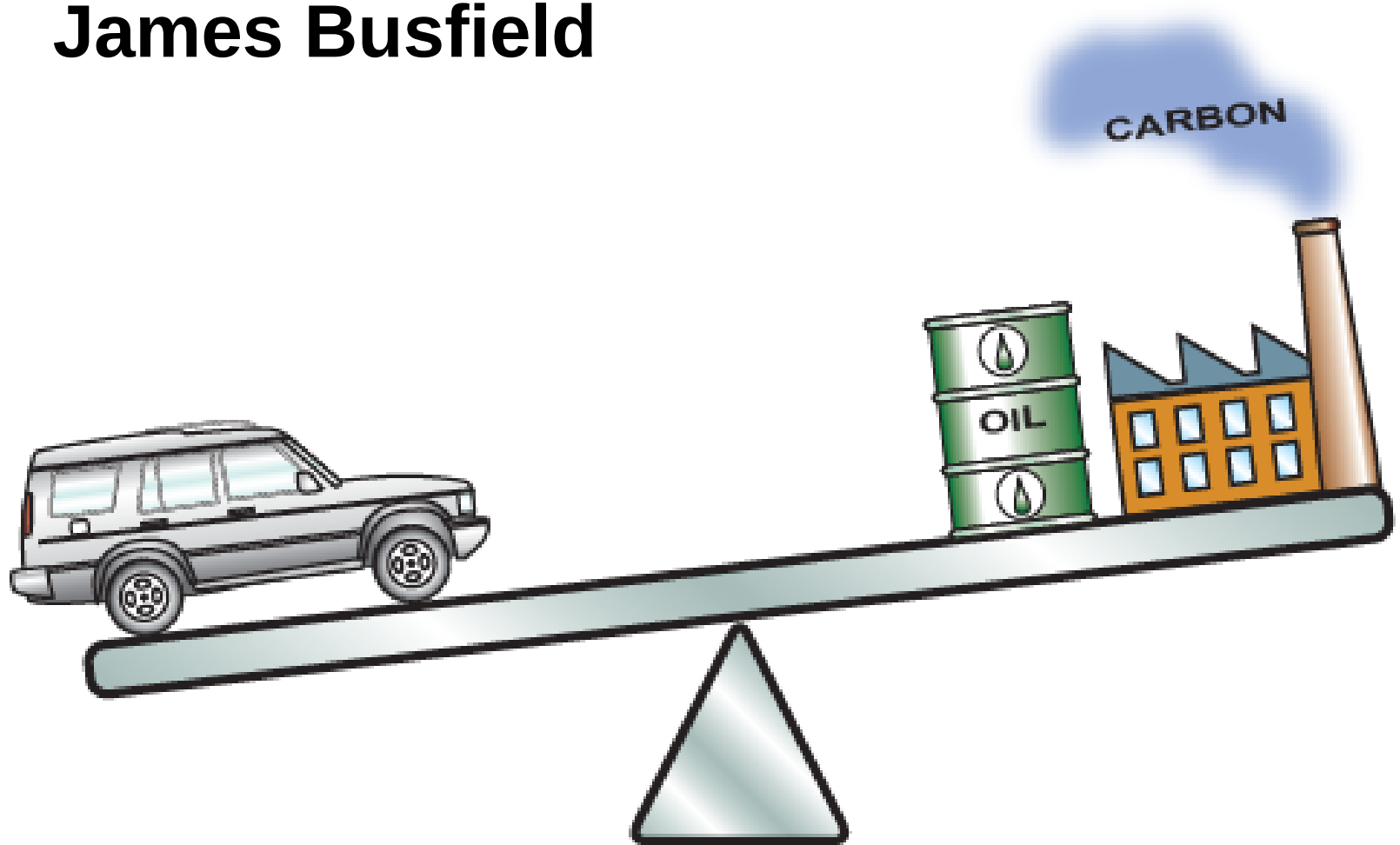


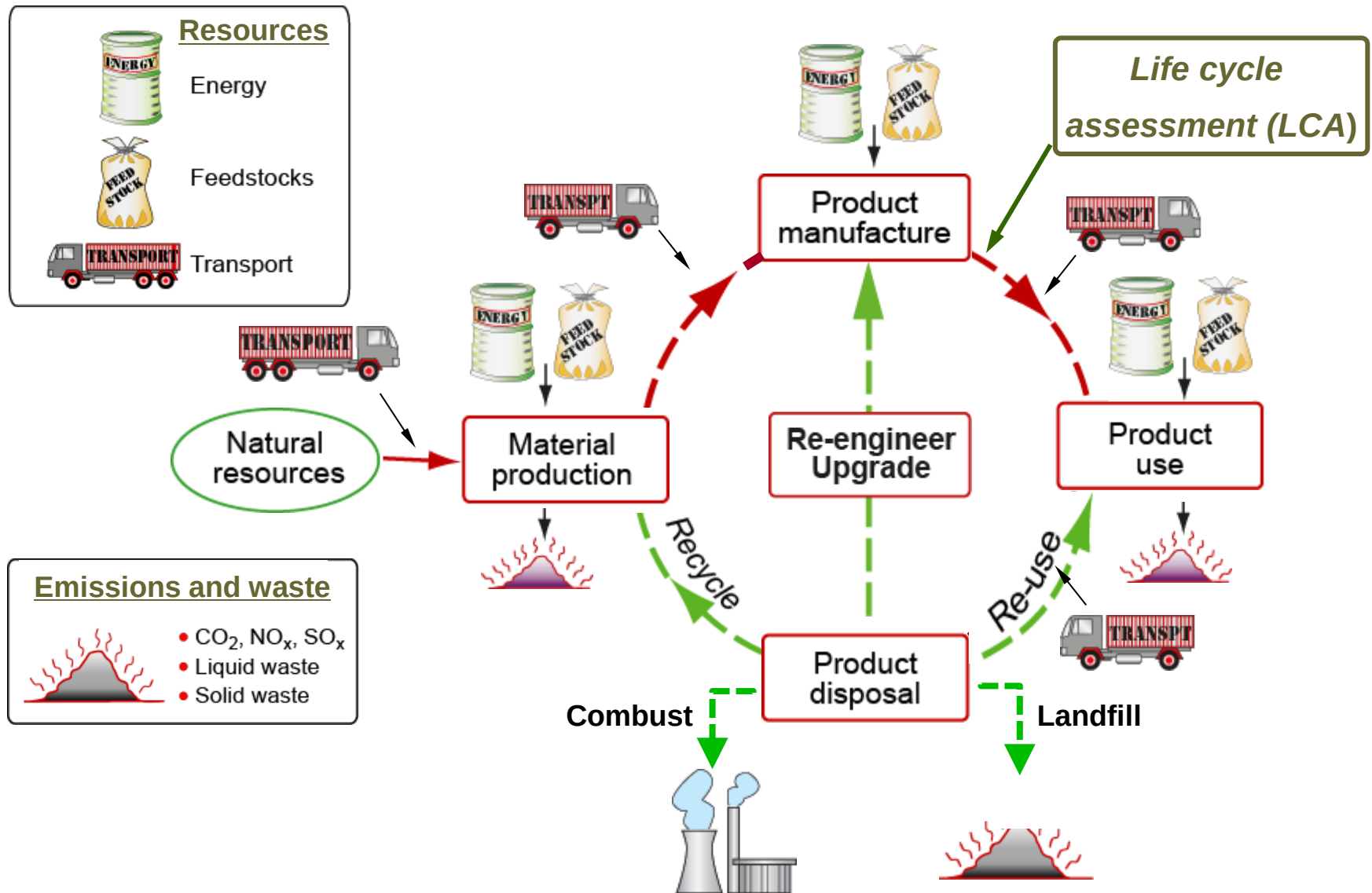
Lecture 9 - Eco Design

Considering the whole life cycle
analysis of a product

James Busfield



The product life-cycle



Life cycle assessment (LCA)

ISO 14040 series PAS 2050

Resource
consumption

Emissions
inventory

Impact
assessment

Typical LCA output

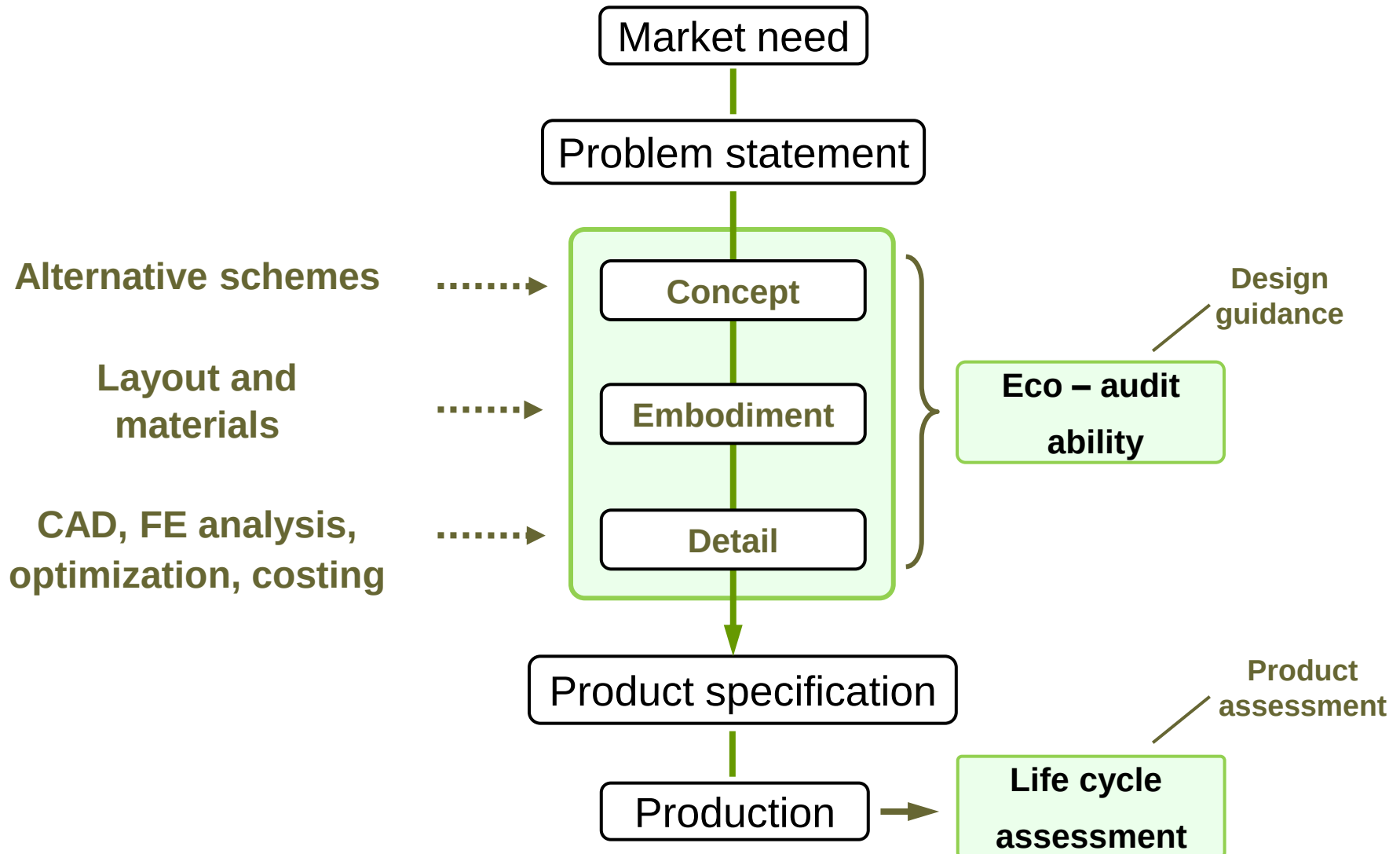
Aluminum cans, per 1000 units

• Bauxite	59	kg
• Oil fuels	148	MJ
• Electricity	1572	MJ
• Energy in feedstock	512	MJ
• Water use	1149	kg
• Emissions: CO ₂	211	kg
• Emissions: CO	0.2	kg
• Emissions: NO _x	1.1	kg
• Emissions: SO _x	1.8	kg
• Particulates	2.47	kg
• Ozone depletion potential	0.2 X 10 ⁻⁹	
• Global warming potential	1.1 X 10 ⁻⁹	
• Acidification potential	0.8 X 10 ⁻⁹	
• Human toxicity potential	0.3 X 10 ⁻⁹	

Roll up into an
“eco-indicator” ?

- Full LCA is **expensive**, and requires great **detail and experience**
- It is also subject to uncertainty

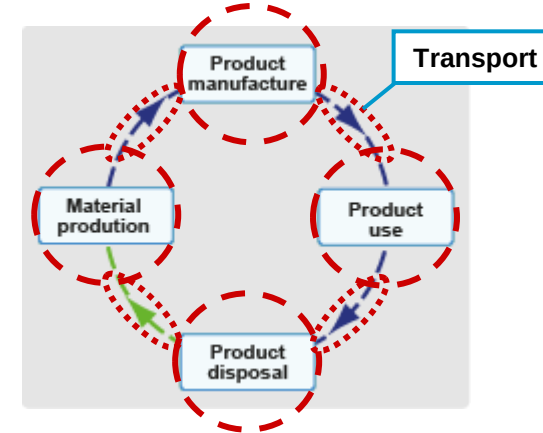
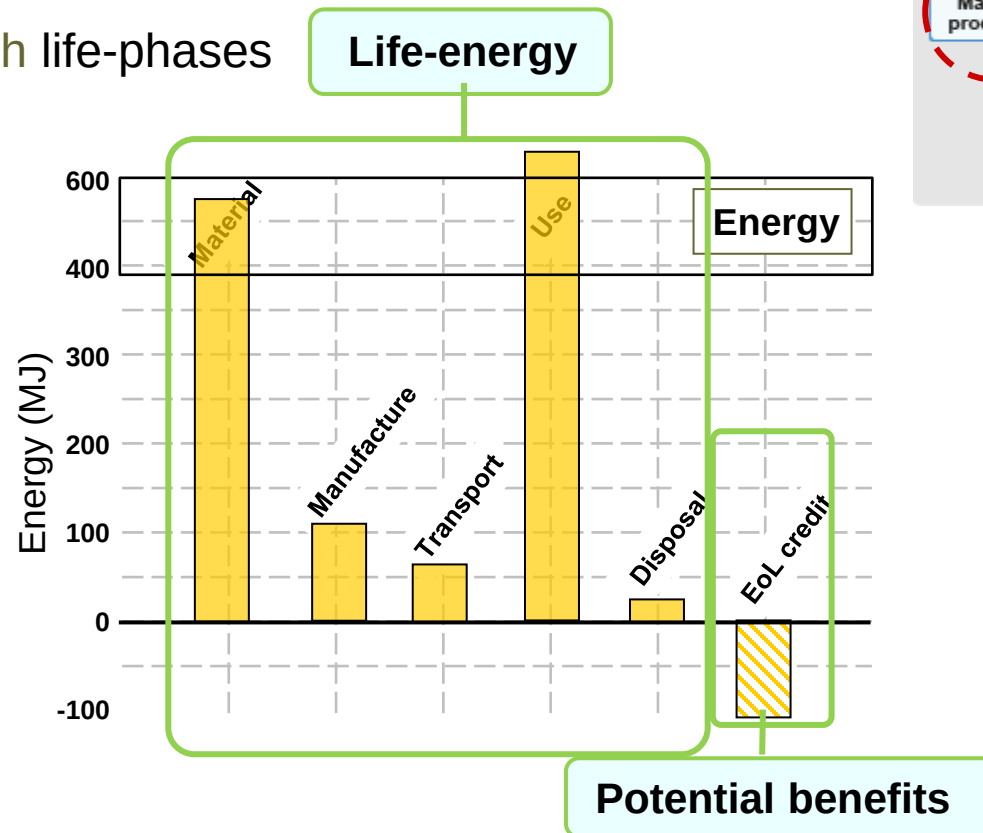
Design Guidance or Product Assessment



Eco-audit for design

Need: Fast **Eco-audit** with sufficient precision to guide decision-making

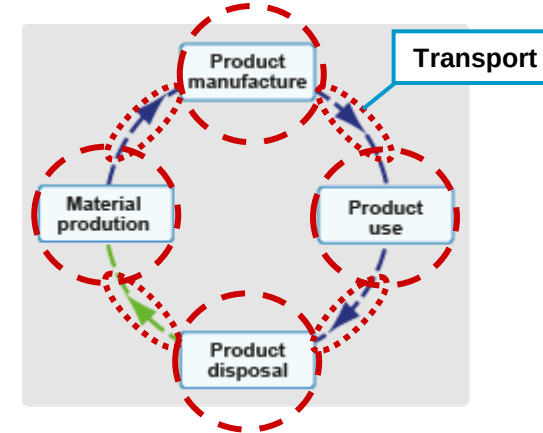
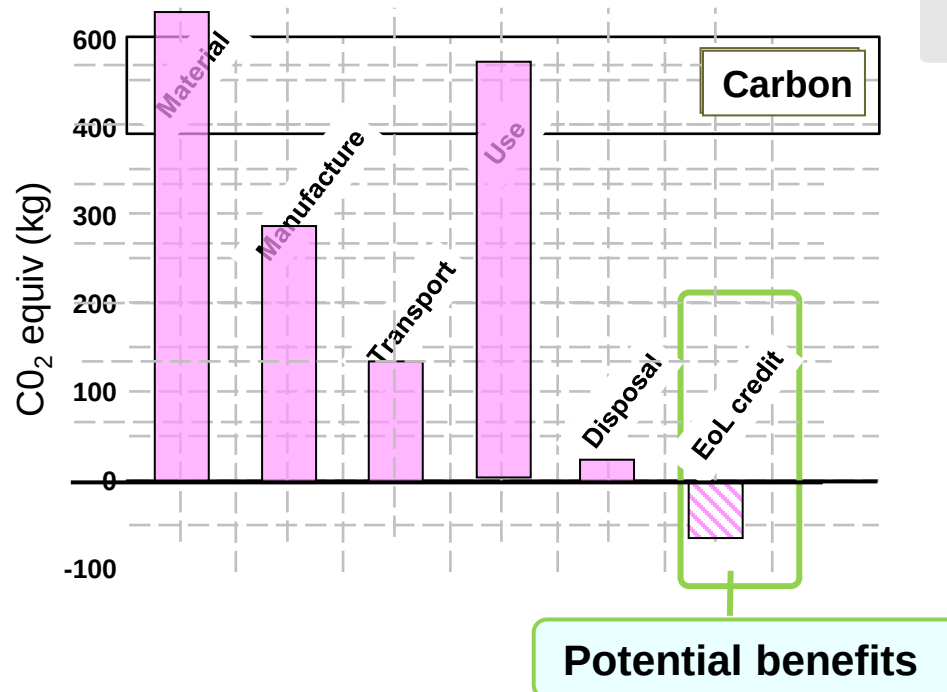
- 1 resource – energy (oil equivalent)
- Distinguish life-phases



Eco-audit for design

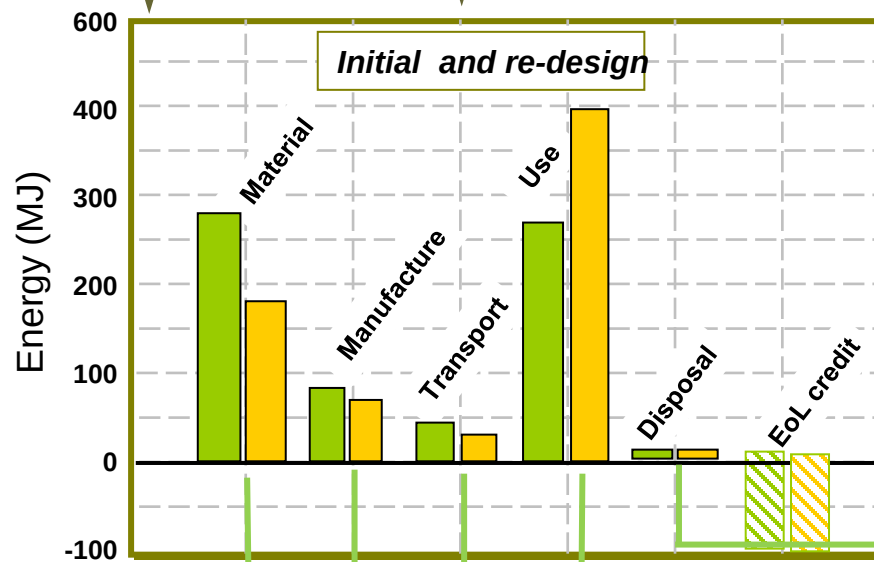
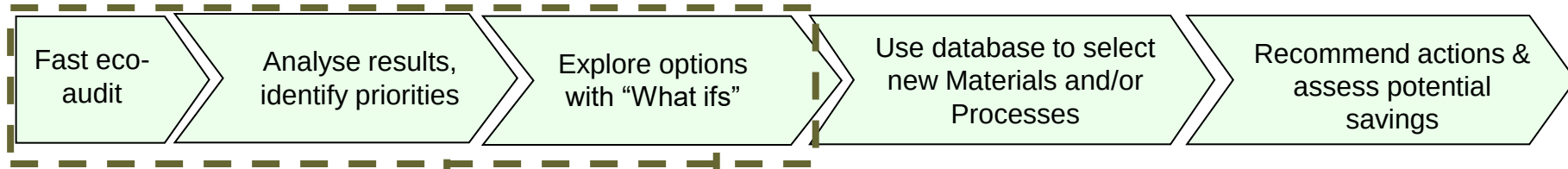
Need: Fast **Eco-audit** with sufficient precision to guide decision-making

- 1 resource – energy (oil equivalent)
- 1 emission – CO₂ equivalent
- Distinguish life-phases



Eco-design: the strategy (1)

The steps



Use eco-audit to identify
Eco-design objective



Eco-audit tools

User inputs

- Bill of materials
- Manufacturing process
- Transport needs
- Duty cycle
- End of life choice



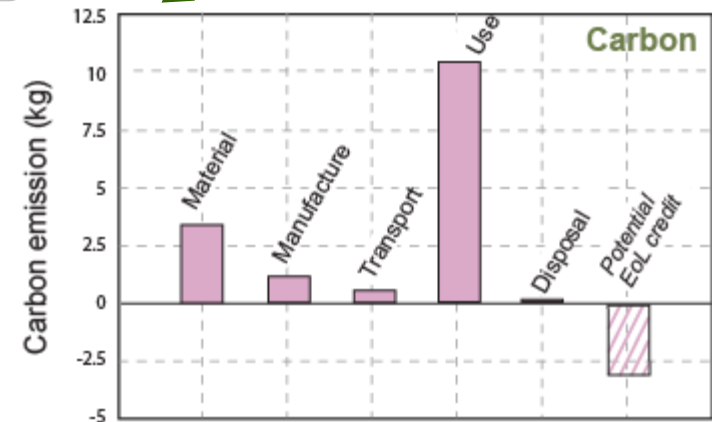
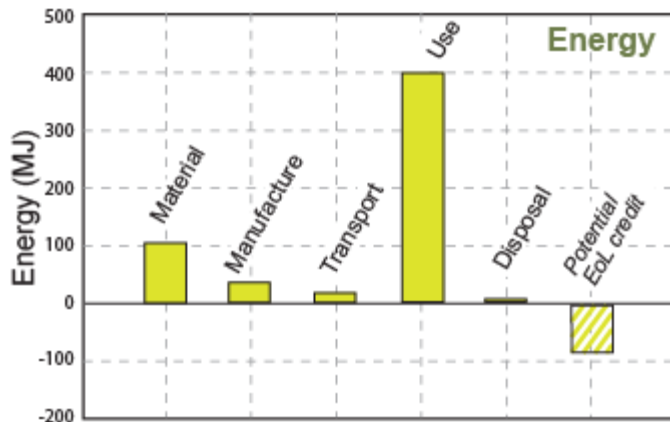
From Database

- Embodied energies
- Process energies
- CO₂ footprints
- Unit transport energies
- Recycling / combustion

Eco audit model

Outputs

(including tabular data)



Typical record showing eco-properties

Database: CES EduPack 2010 Levels 1 & 2

Change...

Table: MaterialUniverse

subset: Edu Level 2 with Eco properties

MaterialUniverse

- Ceramics and glasses
- Hybrids: composites, foams, natural materials
- Metals and alloys
- Polymers and elastomers
 - Elastomers
 - Polymers
 - Thermoplastics
 - Acrylonitrile butadiene styrene (ABS)
 - Cellulose polymers (CA)
 - Ionomer (I)
 - Polyamides (Nylons, PA)
 - Polycarbonate (PC)
 - Polyetheretherketone (PEEK)
 - Polyethylene (PE)
 - Polyethylene terephthalate (PET)**
 - Polyhydroxyalkanoates (PHA, PHB)
 - Poly lactide (PLA)
 - Polymethyl methacrylate (Acrylic, PMMA)
 - Polyoxymethylene (Acetal, POM)
 - Polypropylene (PP)
 - Polystyrene (PS)
 - Polytetrafluoroethylene (Teflon, PTFE)
 - Polyurethane (tpPUR)
 - Polyvinylchloride (tpPVC)
 - Starch-based thermoplastics (TPS)
 - Thermosets

Layout: Edu Level 2 with Eco properties

Show/Hide

Machinability3-4

Weldability5

Geo-economic data for principal component

Annual world production9e6-9.2e6tonne/yr

Reserves* 2.58e8-2.6e8tonne

Primary material production: energy, CO2 and water

Embodied energy, primary production79.6-88MJ/kg

CO2 footprint, primary production2.21-2.45kg/kg

Water usage* 14.7-44.2l/kg

Eco-indicator 95380millipoints/kg

Eco-indicator 99276millipoints/kg

Material processing: energy

Polymer molding energy* 20.4-22.5MJ/kg

Polymer extrusion energy* 7.92-8.73MJ/kg

Polymer machining energy (per unit wt removed)* 2.03-2.24MJ/kg

Material processing: CO2 footprint

Polymer molding CO2* 1.63-1.8kg/kg

Polymer extrusion CO2* 0.633-0.698kg/kg

Polymer machining CO2 (per unit wt removed)* 0.162-0.179kg/kg

Material recycling: energy, CO2 and recycle fraction

Recycle✓

Embodied energy, recycling33.4-37MJ/kg

CO2 footprint, recycling0.928-1.03kg/kg

Recycle fraction in current supply20-22%

Downcycle✓

Combust for energy recovery✓

Heat of combustion (net)* 23-24.2MJ/kg

Combustion CO2* 2.24-2.35kg/kg

Landfill✓

Biodegrade✗

Toxicity ratingNon-toxic

A renewable resource?✗

Environmental notes

The CES EduPack Eco-audit tool

User interface

- Bill of materials
- Manufacturing process
- Transport needs
- Duty cycle
- End of life choice



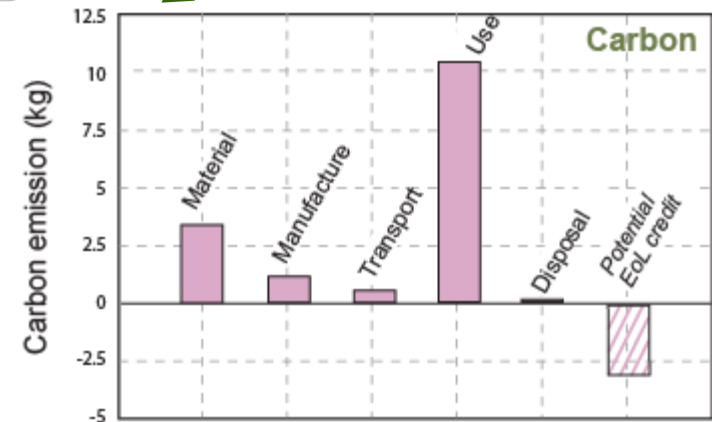
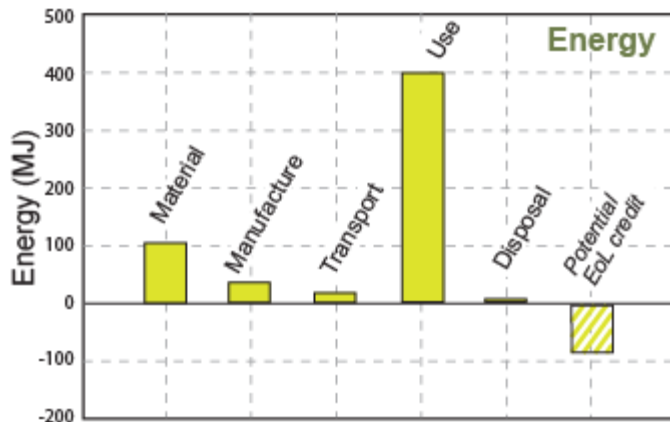
Eco database

- Embodied energies
- Process energies
- CO₂ footprints
- Unit transport energies
- Recycling / combustion

Eco audit model

Outputs

(including tabular data)



Step 1. Material and process energy / CO₂

Component name	Material	Process	Mass (kg)	End of life
Component 1	Aluminum alloys	Casting	2.3	Recycle
Component 2	Polypropylene	Polymer molding	1.85	- Reuse - Refurbish - Recycle - Combust - Landfill
Component 3	Glass	Glass mould	7	

CES EduPack materials tree

- Material Universe
 - Ceramics and glasses
 - Hybrids: composites, foams, natural materials
 - Metals and alloys
 - Ferrous
 - Non-ferrous
 - Aluminum alloys
 - Magnesium alloys
 - Nickel alloys
 - Tin
 - Titanium alloys
 - Tungsten alloys
 - Zinc alloys
 - Polymers and elastomers
 - Elastomers
 - Thermoplastics
 - Acrylonitrile butadi
 - Cellulose polymers
 - Ionomer (I)
 - Polyamides (Nylons)

Energy (MJ)

End of life options

- Reuse
- Refurbish
- Recycle
- Combust
- Landfill

Step 2. Transport

Transport stage

Stage 1

Stage 2

Transport type

32 tonne truck



- Sea freight
- River / Canal freight
- Rail freight
- 32 tonne truck
- 14 tonne truck
- Light goods vehicle
- Air freight - short haul
- Air freight - long haul
- Helicopter (Eurocopter AS 35)

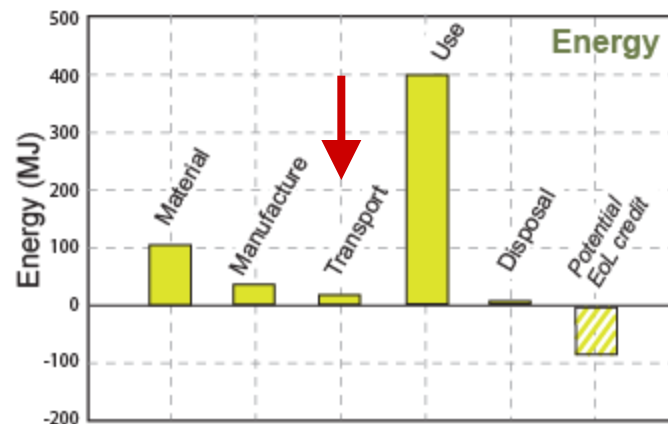
Distance (km)

350

12000

Transport energy

Transport CO₂



Step 3. Use phase – static mode

Energy input and output

Fossil fuel to electric

Power rating

1.2

kW

Usage

365

days per year

Usage

0.5

hours per day

Total energy and CO₂ for use

Energy conversion path

Fossil fuel to heat, enclosed system

Fossil fuel to heat, vented system

Fossil fuel to electric

Electric to mechanical

Electric to heat

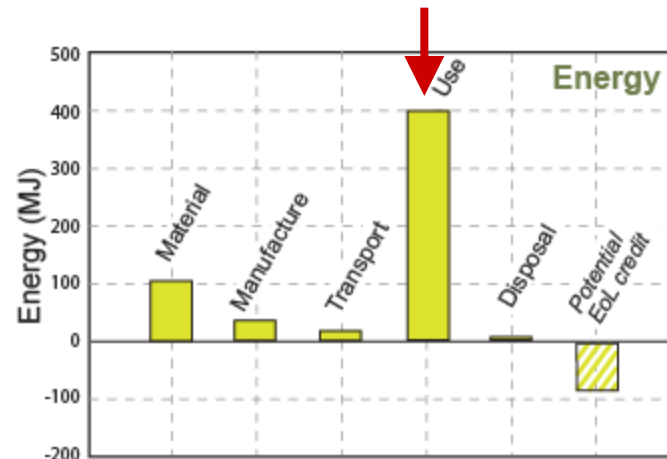
Electric to mechanical (electric motor)

Electric to chemical (lead-acid battery)

Electric to chemical (Lithium-ion battery)

Electric to light (incandescent lamp)

Electric to light (LED)



Bottled water (100 units)



- 1 litre PET bottle with PP cap
- Blow moulded
- Filled in France, transported 550km to UK
- Refrigerated for 2 days, then drunk

Product name: *PET bottle*

Compare with... ▼

Clear

Open

Save

Number	Name	Material	Process	Mass (kg)	End of life
<i>100</i>	<i>Bottles</i>	<i>PET</i> ▼	<i>Moulding</i> ▼	<i>0.04</i>	<i>Recycle</i> ▼
<i>100</i>	<i>Caps</i>	<i>Polyprop</i> ▼	<i>Moulding</i> ▼	<i>0.001</i>	<i>Landfill</i> ▼
<i>100</i>	<i>Water</i>	▼	▼	<i>1.0</i>	▼

Transport

Stage 1

14 tonne truck ▼

550 km

Survey charts

Use - refrigeration

Fossil to electric

0.12 kW ▼

2 days

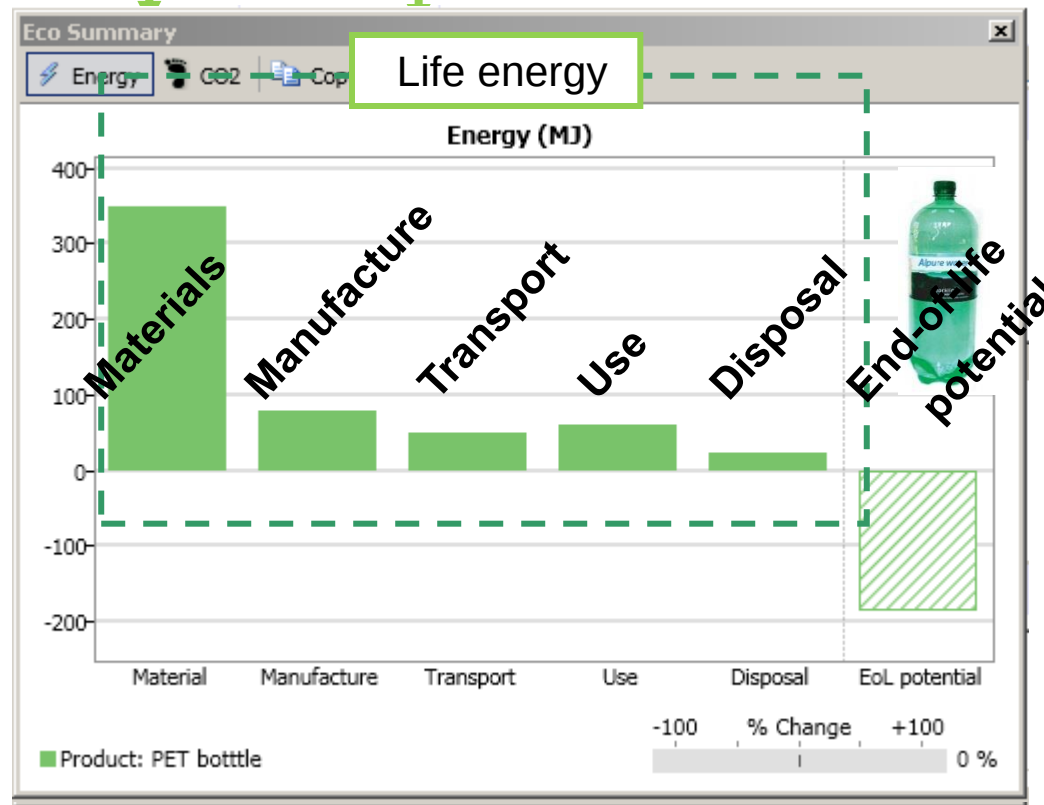
24 hrs/day

Full report

Output: summary charts

Toggle between energy and carbon footprint

Copy or print the chart



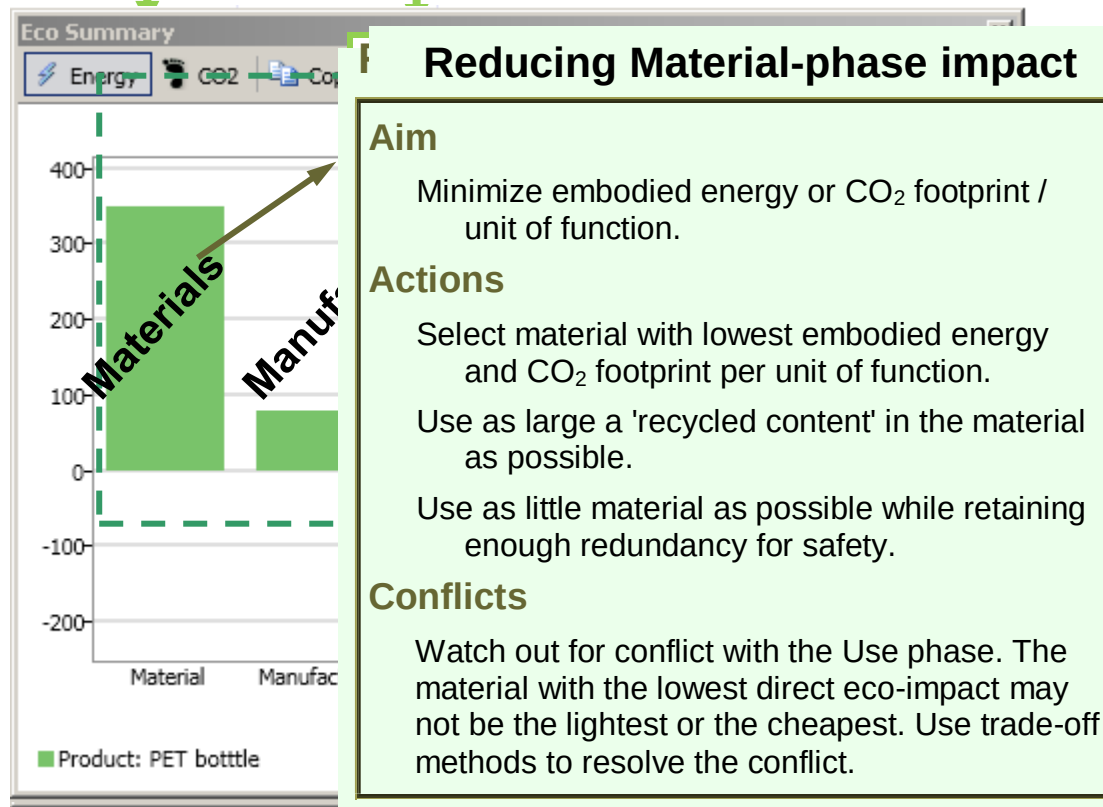
Which phase has the largest impact?

Materials

Output: summary charts

Toggle between energy and carbon footprint

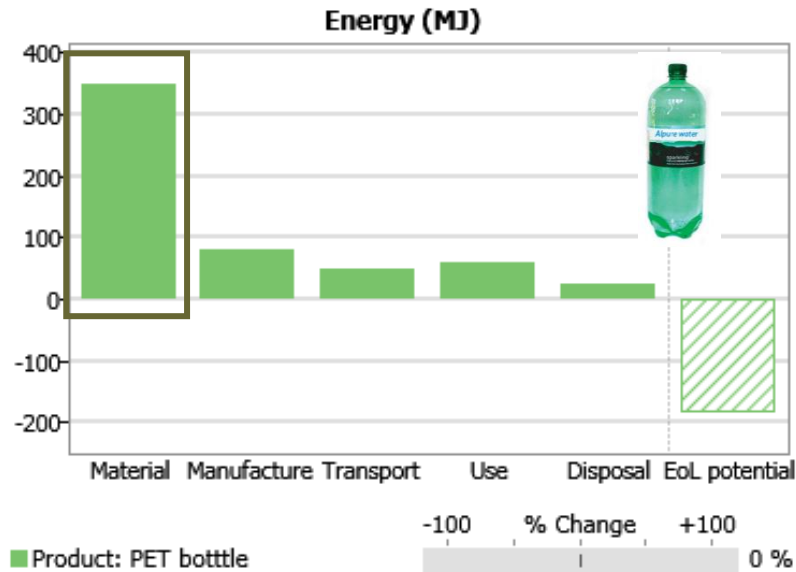
Copy *or* print the chart



Which phase has the largest impact?

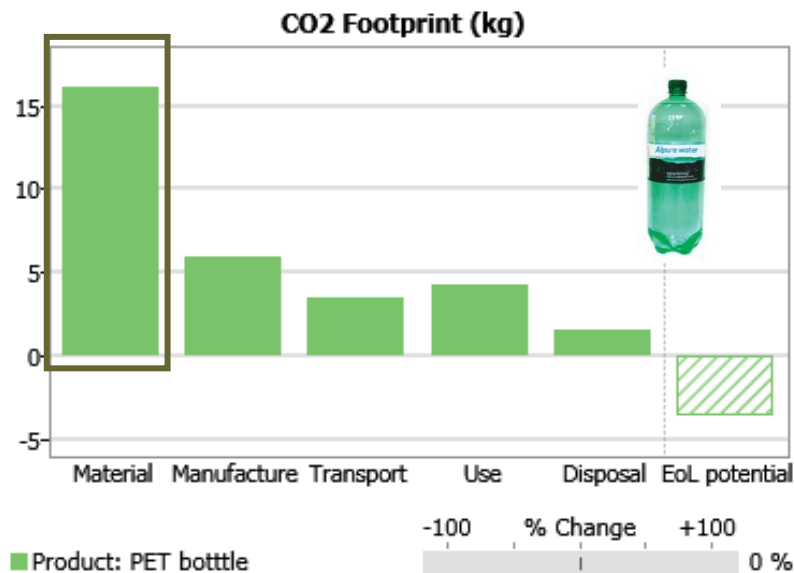
Materials

Actions and comparisons



The audit reveals the most energy and carbon intensive steps...

... and allows rapid "What if..."



PET



Glass ?

Change the materials



- 1 litre **glass bottle** with **aluminum cap**
- **Glass moulded**
- Filled in France, transported 550km to UK
- Refrigerated for 2 days, then drunk

Product name: **Glass bottle**

Compare with...

Clear

Open

Save

Number	Name	Material	Mass (kg)	End of life
100	Bottles	Soda glass	0.45	Recycle
100	Caps	Aluminum	0.002	Landfill
100	Water		1.0	

Copy of current project

New project

Saved project

Transport

Stage 1

14 tonne truck

550 km

Use - refrigeration

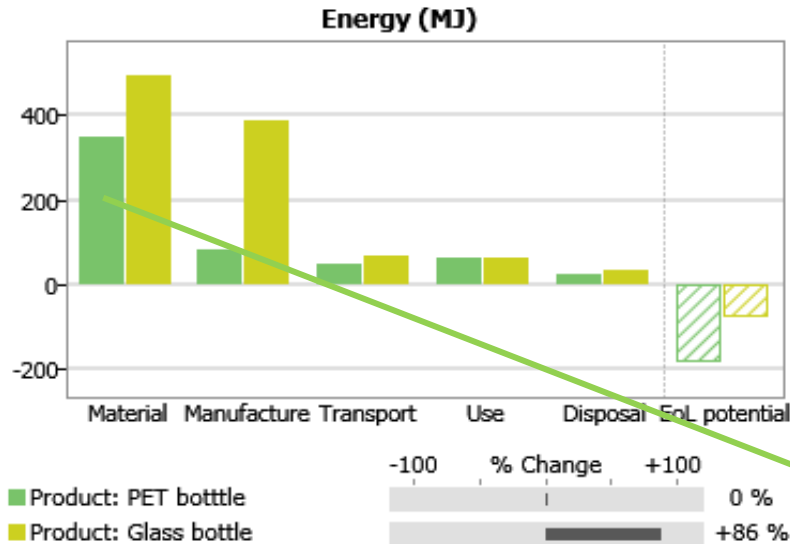
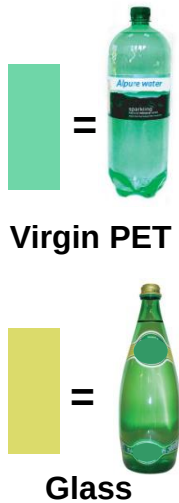
Fossil to electric

0.12 kW

2 days

24 hrs/day

Actions and comparisons



The fast comparison allows design decisions on-the-fly

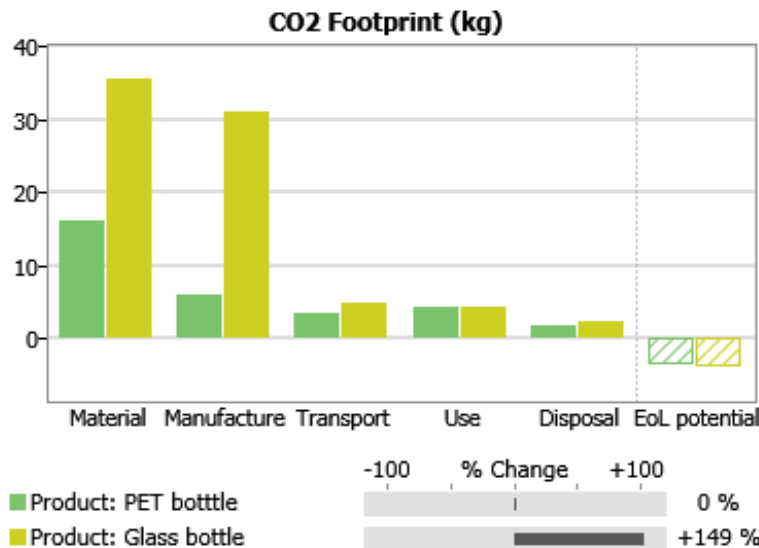
Reducing impact

Actions

- Use as large a 'recycled content' in the material as possible.

What if.....

100% recycled PET?



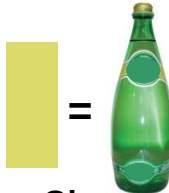
Actions and comparisons

Click
Compare with....
Copy of current
content

Set Recycle
content to 100%



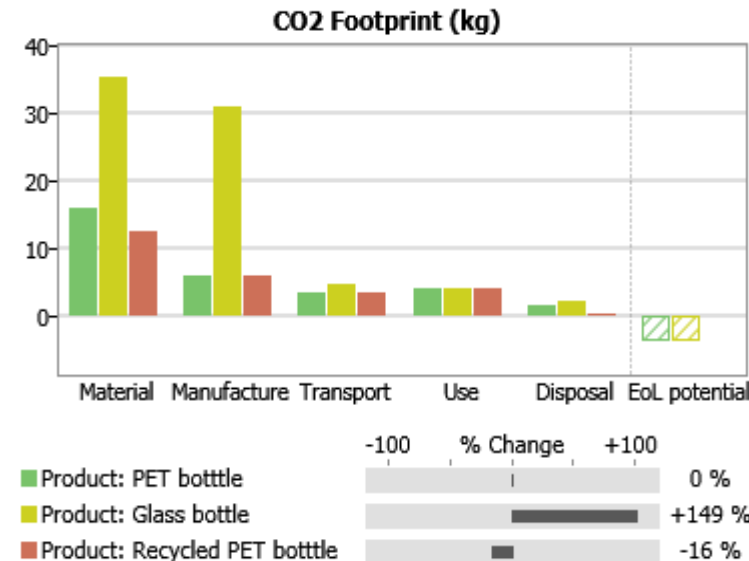
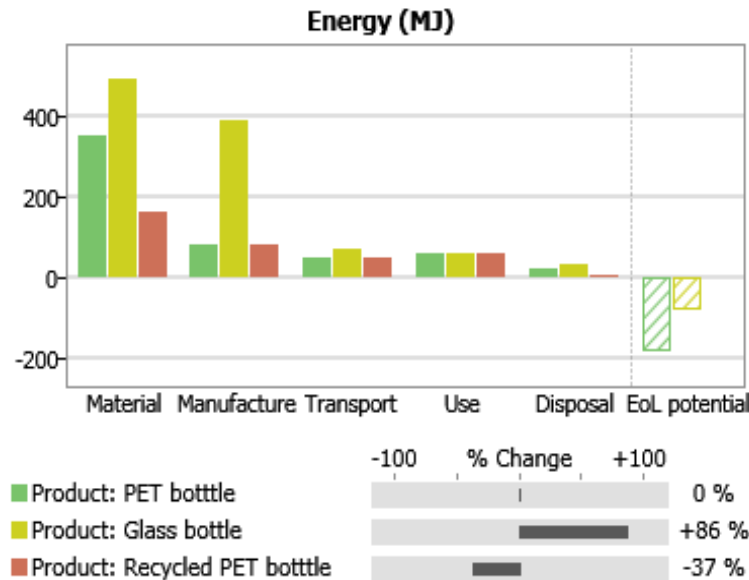
Virgin PET



Glass



Recycled PET



Can explore:

- Material choice
- Recycle content
- Transport mode
- Transport distance
- Use pattern
- Electric energy mix
- End of life choice

Many projects available
as project files.

Electric kettle

Bill of materials and processes



2 kW electric kettle

- Made SE Asia
- Air freight to UK
- Life: 3 years

Qty.	Component name	Material	Recycled content	Mass (kg)	Primary process	End of life
1	Kettle body	 Polypropylene (PP)	Virgin (0%)	0.86	Polymer molding	Recycle
1	Heating element	 Nickel-chromium alloys	Virgin (0%)	0.026	Wire drawing	Recycle
1	Casing, heating element	 Stainless steel	Virgin (0%)	0.09	Rough rolling, forging	Recycle
1	Internal insulation	 Alumina	Virgin (0%)	0.03	Incl. in material value	Landfill
1	Thermostat	 Copper alloys	Virgin (0%)	0.02	Rough rolling, forging	Recycle
1	Plug body	 Phenolics	Virgin (0%)	0.037	Polymer molding	Landfill
1	Plug pins	 Brass	Virgin (0%)	0.03	Extrusion, foil rolling	Recycle
1	Cable sheath, 1 meter	 Natural rubber (NR)	Virgin (0%)	0.06	Polymer molding	Landfill
1	Cable core, 1 meter	 Copper	Virgin (0%)	0.015	Wire drawing	Recycle
1	Packaging, foam	 Rigid Polymer Foam (MD)	Virgin (0%)	0.015	Polymer molding	Landfill
1	Packaging, cardboard	 Paper and cardboard	Virgin (0%)	0.125	Incl. in material value	Recycle
1	Residual components	 Polycarbonate (PC)	Virgin (0%)	0.04	Polymer molding	Downcycle

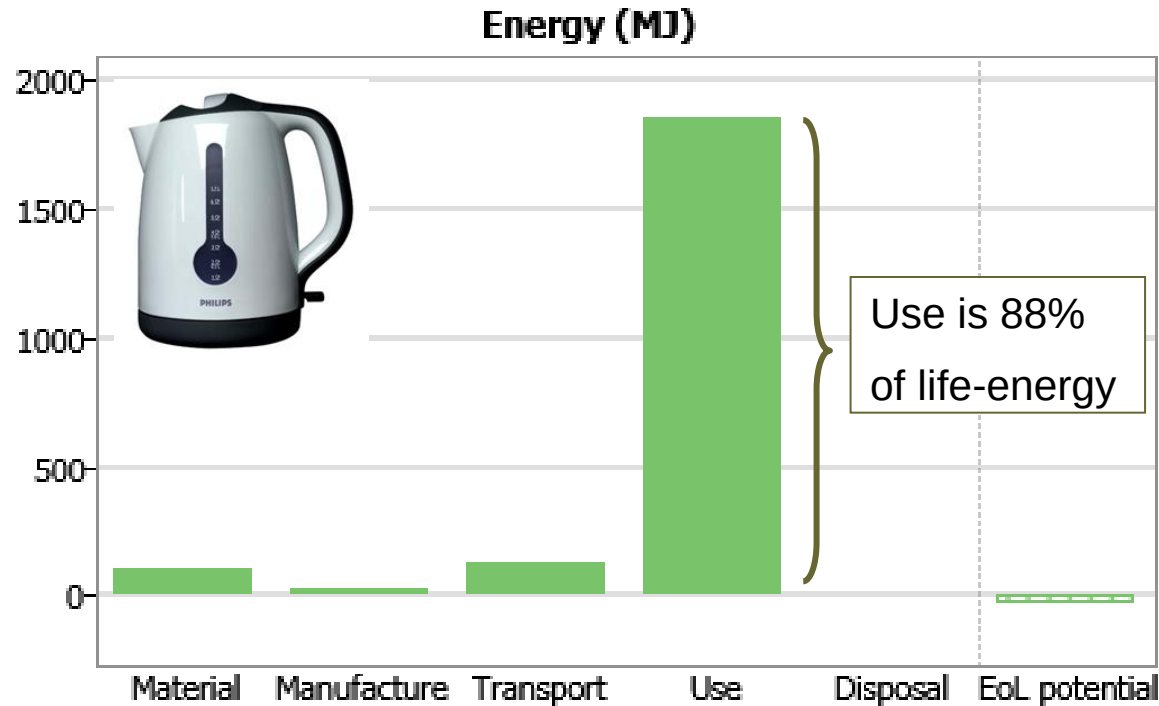
Transport

- 12,000 km, air freight
- 250 km 14 tonne truck

Use

- 6 minutes per day
- 300 days per year
- 3 years

Eco audit: the jug kettle



What do we learn?

- Little gained by change of material for its own sake
- Much gained by insulation – double wall with foam or vacuum
- Or seek new concept: hot water on the fly – only as much as needed